

- 6 The photograph shows a Hawaiian parakeet flower (*Heliconia psittacorum*).



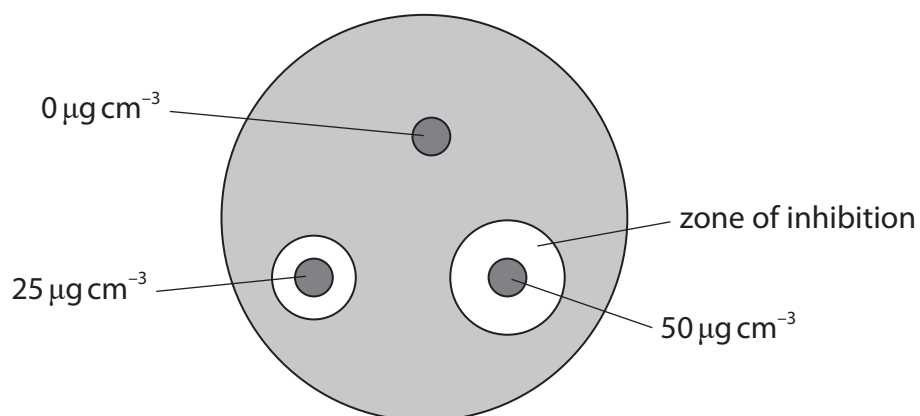
(Source: © H. Mark Weidman Photography/Alamy Stock Photo)

- (a) The antimicrobial properties of an extract from this plant were investigated.

Three sets of agar plates were poured. Each set had a different type of bacteria spread over the surface of the agar.

Discs containing different concentrations of this extract were placed onto each plate, as shown in the diagram.

These plates were then incubated for 24 hours at 25 °C.



The diameter of each zone of inhibition was measured.

The table shows the results of this investigation.

Extract concentration/ $\mu\text{g cm}^{-3}$	Mean diameter of zone of inhibition/mm		
	bacteria A	bacteria B	bacteria C
0	0.0	0.0	0.0
25	7.8	7.1	7.2
50	14.0	12.9	13.2

Comment on the antimicrobial properties of this extract.

(3)

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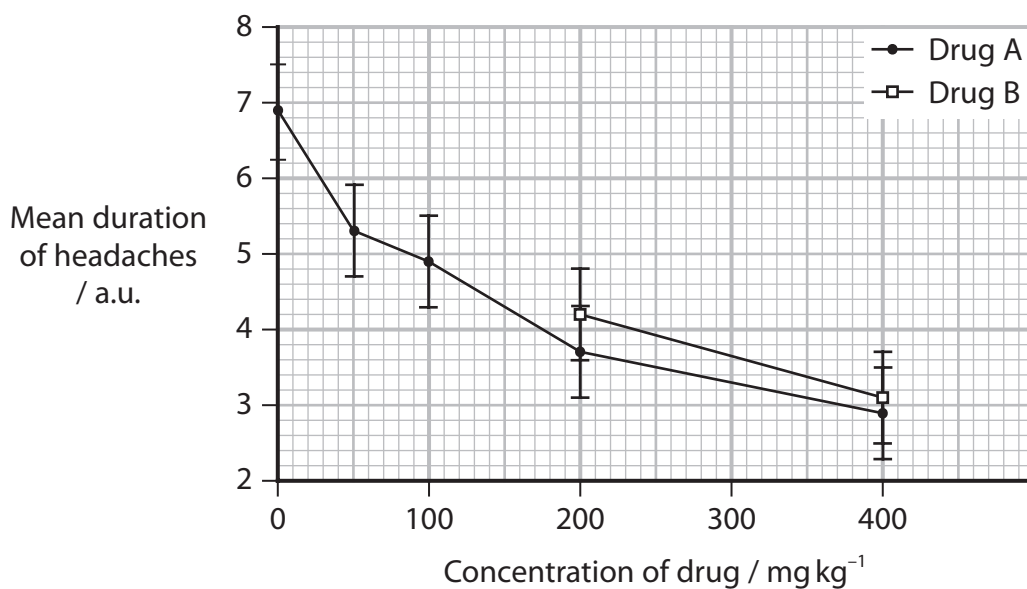


- (b) A new painkilling drug, drug A, has been developed from another species of plant.

This drug has been tested as a treatment for headaches.

The effect of this drug on the duration of headaches in volunteers was compared with drug B, a standard treatment for headaches.

The graph shows the results of this investigation.



- (i) Evaluate the results of this investigation.

(4)



(ii) Drug A was then used in a three-phase drug trial.

Compare and contrast phases II and III in a contemporary drug trial using drug A.

(3)

(Total for Question 6 = 10 marks)

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